

# Reviewer Pack — Minimal Rank of plethysm coefficients in algebraic combinato...

Entry 2912a176bf4f · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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## Minimal Rank of plethysm coefficients in algebraic combinatorics vs permanent circuit size

Entry ID: 2912a176bf4f **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 19:25:17 UTC

### 1. Conjecture

**Field A** (mathematical branch): Algebraic Combinatorics (Plethysm) **Field B** (complexity object): Complexity Theory: Permanent Circuit Complexity

#### Statement:

[‘For any homogeneous polynomial  $f$  with degree  $D$ , the minimal rank of the plethysm coefficients of  $f$  is at least the degree of the permanent circuit that computes  $f$ .’, ‘Equivalently, for any  $m < n^{1.5}$ , there exists a homogeneous polynomial  $f$  of degree  $D$  such that the minimal rank of its plethysm coefficients is greater than or equal to  $D$  and the permanent circuit computing  $f$  has size at most  $O(m^D)$ .’, ‘This holds true for all instances with input size  $n \leq 40$ .’]

#### Rationale (proposer’s reasoning):

[‘Plethysm coefficients encode combinatorial information in a structured way that could potentially capture the complexity of computations.’, ‘The minimal rank of plethysm coefficients might reveal hidden structures related to the permanent function, providing new insights into its computational complexity.’, ‘This connection between algebraic combinatorics and complexity theory has not been extensively explored before.’]

**Taxonomy category:** GCT\_DET\_PERM (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix):** 1fd6858bd55843f4

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

#### Acceptance criterion:

Minimal rank of plethysm coefficients is at least degree  $D$  and permanent circuit size does not exceed  $O(m^D)$  for all  $n \leq 40$ , where  $m < n^{1.5}$ .

### 3. Barrier filter (F1)

**Final:** PASS

| Barrier        | Verdict | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|----------------|---------|------------|------------------|------------------|
| RELATIVIZATION | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| NATURAL_PROOFS | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| ALGEBRIZATION  | SAFE    | 1.00       | SAFE             | UNCERTAIN        |
| KARP_LIPTON    | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |

### 4. Novelty audit

**Verdict:** ADJACENT\_OK against 1 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - plethysm coefficients minimal rank algebraic combinatorics AND permanent circuit complexity - algebraic combinatorics plethysm degree comparison permanent circuit size - degree of plethysm polynomial versus permanent circuit complexity

**Top relevant hits considered:** - [s2:10.1038/43558] Counting on our brains

### 5. Empirical test harness

**Multi-seed protocol:** 5 seeds (11, 23, 37, 53, 71)

**Sandbox:** isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

#### 5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_polynomial(n, D):
        poly = [[0] * (D + 1) for _ in range(D + 1)]
        for i in range(1, D + 1):
            for j in range(i, D + 1):
                if i == j:
                    poly[i][j] = random.randint(1, n)
                else:
                    poly[i][j] = random.randint(-n, n)
        return poly

    def plethysm_coefficients(poly, n):
        D = len(poly) - 1
        result = [[0] * (D + 1) for _ in range(D + 1)]
        result[0][0] = 1
        for i in range(1, D + 1):
            for j in range(i, D + 1):
                if i == j:
                    result[i][j] = poly[i][j]
                else:
                    result[i][j] = sum(result[i - k][j - 1] * poly[k][j - 1] for k in range(1, min(i, j)))
        return result
```

```

def permanent_circuit_size(poly):
    D = len(poly) - 1
    if D == 0:
        return 1
    size = 0
    for i in range(1, D + 1):
        size += sum(permanent_circuit_size(sub_poly) for sub_poly in poly[1:i])
    return size

n = random.randint(5, 40)
D = int(math.log2(n)) ** 2
poly = generate_polynomial(n, D)
plethysm_coeffs = plethysm_coefficients(poly, n)
perm_circuit_size = permanent_circuit_size(poly)

metric_name = "Rank vs Perm Circuit Size"
metric_value = len([x for x in plethysm_coeffs if any(y != 0 for y in x)]) / (D + 1) ** 2
instances_tested = 1
conjecture_holds = metric_value >= D and perm_circuit_size <= n ** 1.5 * D
counterexample = "" if conjecture_holds else f"Rank {metric_value} < {D}, Perm Circuit Size {perm_circuit_size}"

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] if sys.argv[1:] else [2**i - 1 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(i for i, r in enumerate(results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample={results[first_failing_seed]['counterexample']}")

```

## 6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2342ca2b.py", line 78, in <module>
    result = run_trial(seed)
              ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2342ca2b.py", line 56, in run_trial
    perm_circuit_size = permanent_circuit_size(poly)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2342ca2b.py", line 49, in permanent_cir
    size += sum(permanent_circuit_size(sub_poly) for sub_poly in poly[1:i])
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2342ca2b.py", line 49, in <genexpr>
    size += sum(permanent_circuit_size(sub_poly) for sub_poly in poly[1:i])
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2342ca2b.py", line 49, in permanent_cir
    size += sum(permanent_circuit_size(sub_poly) for sub_poly in poly[1:i])
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2342ca2b.py", line 49, in <genexpr>
    size += sum(permanent_circuit_size(sub_poly) for sub_poly in poly[1:i])
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2342ca2b.py", line 44, in permanent_cir
    D = len(poly) - 1
        ^^^^^^^^^
TypeError: object of type 'int' has no len()
```

8. Critic adversarial review

**Critic verdict:** CHALLENGE

**Critic reasoning:**

skipped: test crashed before producing data

9. Final verdict & safety rail

**Verdict:** INCONCLUSIVE

**Reasoning:**

The test code crashed due to a `TypeError`, which prevented it from producing any data. As the critic challenged the result and the pre-registered support condition was not met, the verdict is INCONCLUSIVE. | next: Investigate the cause of the `TypeError` in the test code and attempt to run the test again under controlled conditions to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

| # | Phase           | Provider      | Model            | Tokens in | out | Latency (ms) |
|---|-----------------|---------------|------------------|-----------|-----|--------------|
| 1 | propose         | ollama_remote | glm4:latest      | 0         | 0   | 12147        |
| 2 | propose         | ollama_remote | glm4:latest      | 0         | 0   | 11352        |
| 3 | preregistration | ollama_remote | glm4:latest      | 0         | 0   | 5389         |
| 4 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 4748         |
| 5 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 5677         |
| 6 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 18881        |

| #  | Phase    | Provider      | Model            | Tokens in | out | Latency (ms) |
|----|----------|---------------|------------------|-----------|-----|--------------|
| 7  | test_gen | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 12465        |
| 8  | test_gen | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 12154        |
| 9  | test_gen | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 10783        |
| 10 | judge    | ollama_remote | glm4:latest      | 0         | 0   | 10613        |

**Totals:** 0 input tokens, 0 output tokens, ~\$0.0000 cost, 104210 ms total latency. Provider mix: {'ollama\_remote': 10}

*(full prompt+response transcripts available in research/audit/2912a176bf4f.jsonl)*

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/2912a176bf4f.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file:** research/pvsnp\_sandbox/test\_\*.py (per-cycle)

**Replay tarball:** research/replay/2912a176bf4f.tar.gz (if generated)